

Abstract Algebra

- A Journey from Concretization to Abstraction -

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A document presented for
the Abstract Algebra

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Contents

- 1 Visual Group Theory 3
- 2 A Mapping $\mathbb{C}[x] \rightarrow \mathbb{C}$ 10
 - 2.1 Observation 10
 - 2.2 Zorn's Lemma and Basis 20
 - 2.2.1 Relations 20
- 3 Localization 23
 - 3.1 For completeness: The formal definition/statement 24

Chapter 1

Visual Group Theory

Mathematical Definitions and Examples

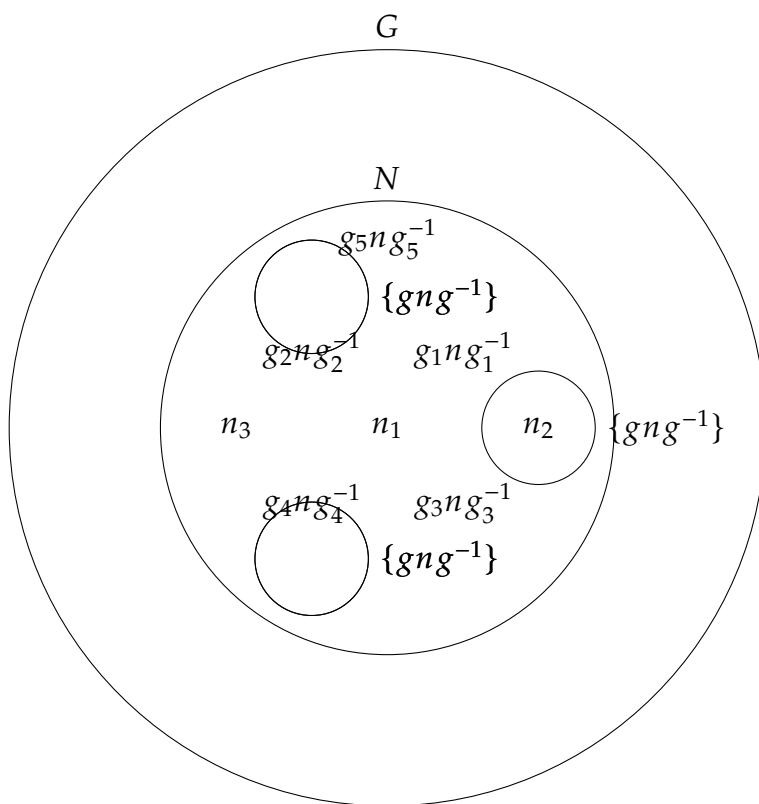
1. Normal Subgroup

Definition:

A subgroup N of a group G is called a *normal subgroup* if it is invariant under conjugation, that is, for every element $g \in G$ and $n \in N$, the element $gng^{-1} \in N$.

Symbolically:

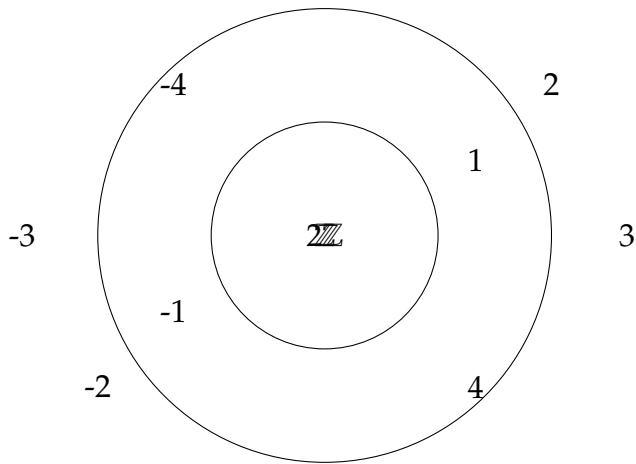
$$N \triangleleft G \iff \forall g \in G, \forall n \in N, gng^{-1} \in N.$$



Examples:

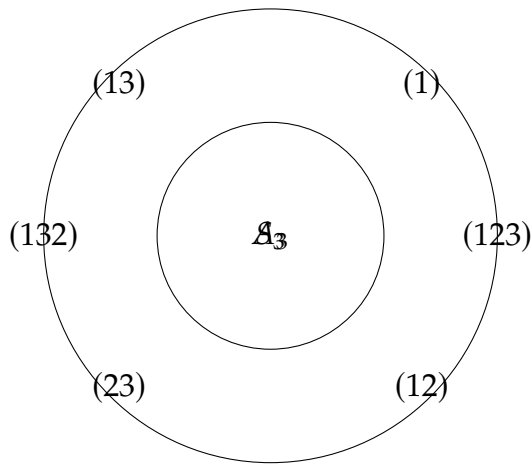
1. In the group of integers $(\mathbb{Z}, +)$, the subgroup $2\mathbb{Z} = \{2k \mid k \in \mathbb{Z}\}$ is a normal subgroup since $\mathbb{Z}$ is abelian.

$$2\mathbb{Z} = \{2k \mid k \in \mathbb{Z}\} \triangleleft \mathbb{Z}$$



2. In the symmetric group S_3 , the subgroup $A_3 = \{(1), (123), (132)\}$ (the alternating group) is a normal subgroup.

$$A_3 = \{(1), (123), (132)\} \triangleleft S_3$$



2. Quotient Group (Factor Group)

Definition:

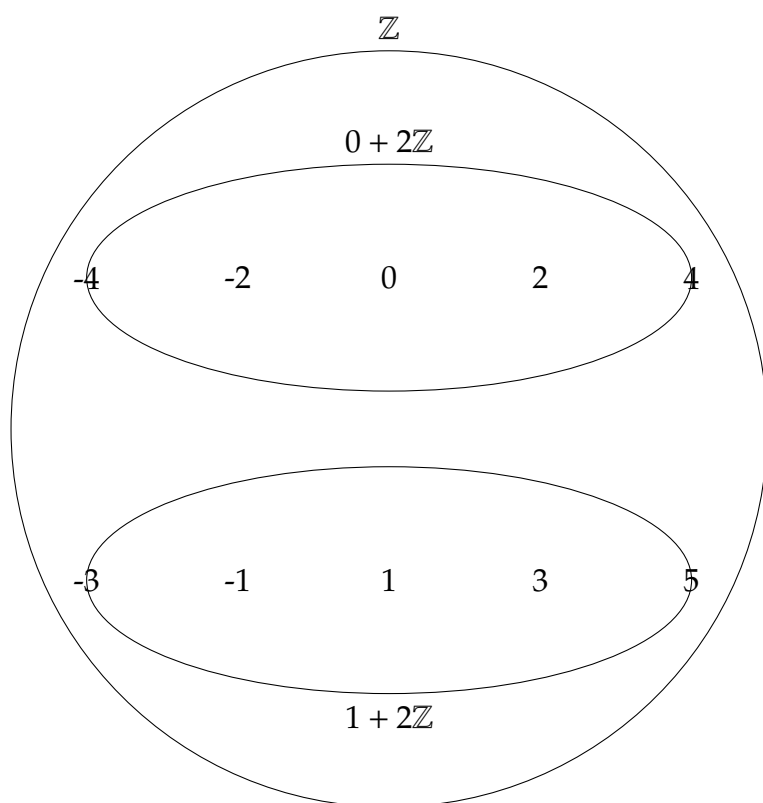
Given a group G and a normal subgroup N of G , the *quotient group* (or *factor group*) G/N is the set of left cosets of N in G with the group operation defined by:

$$(gN) \cdot (hN) = (gh)N$$

Examples:

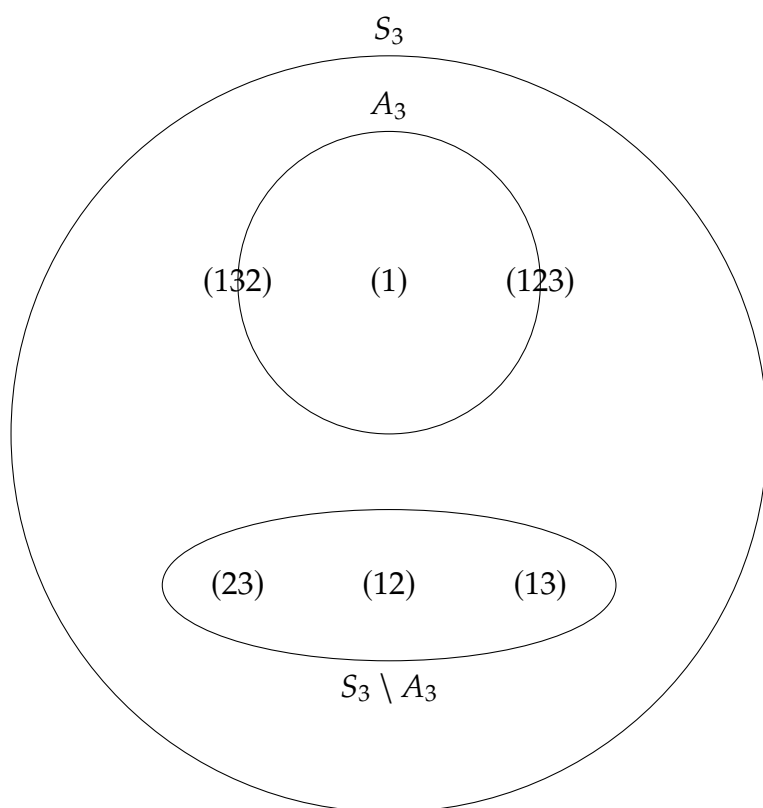
1. For $(\mathbb{Z}, +)$ and $N = 2\mathbb{Z}$, the quotient group $\mathbb{Z}/2\mathbb{Z}$ consists of two cosets: $0 + 2\mathbb{Z}$ and $1 + 2\mathbb{Z}$.

$$\mathbb{Z}/2\mathbb{Z} = \{0 + 2\mathbb{Z}, 1 + 2\mathbb{Z}\}$$



2. In S_3 , the quotient group S_3/A_3 is isomorphic to $\mathbb{Z}_2$.

$$S_3/A_3 \cong \mathbb{Z}_2$$



3. Ring

Definition:

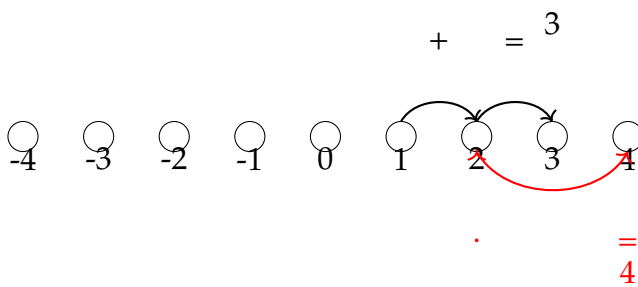
A *ring* is a set R equipped with two binary operations $+$ and $\cdot$ (addition and multiplication) such that:

1. $(R, +)$ is an abelian group.
2. $(R, \cdot)$ is a monoid.
3. Multiplication is distributive over addition: for all $a, b, c \in R$, $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.

Examples:

1. The set of integers $\mathbb{Z}$ with usual addition and multiplication.

$\mathbb{Z}$ is a ring



2. The set of 2×2 matrices over $\mathbb{R}$, $M_2(\mathbb{R})$, with matrix addition and multiplication.

$M_2(\mathbb{R})$ is a ring

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \cdot \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

4. Ideal

Definition:

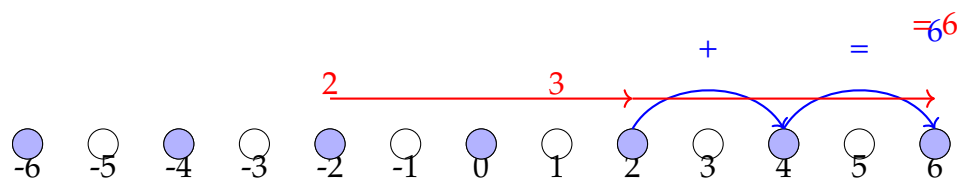
An *ideal* I of a ring R is a subset of R such that:

1. I is an additive subgroup of R .
2. For every $r \in R$ and $i \in I$, both ri and ir are in I .

Examples:

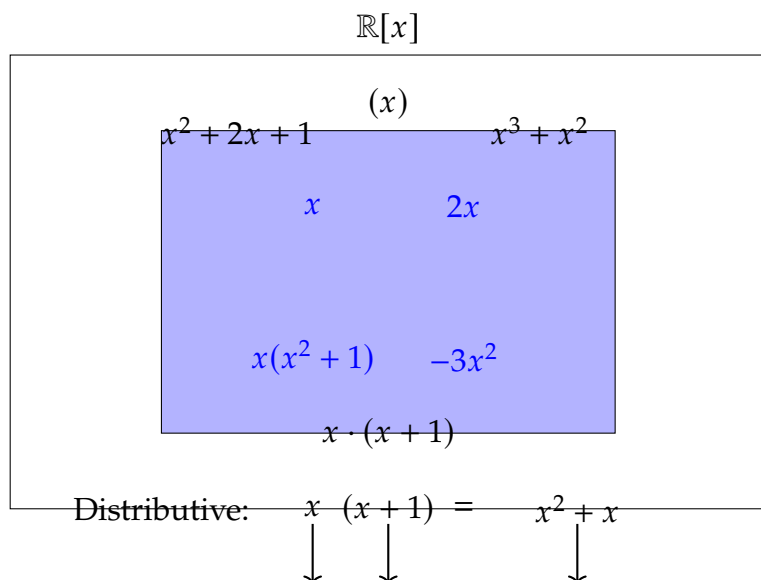
1. In $\mathbb{Z}$, the set $2\mathbb{Z}$ (all even integers) is an ideal.

$$2\mathbb{Z} = \{2k \mid k \in \mathbb{Z}\} \text{ is an ideal of } \mathbb{Z}$$



2. In the ring $\mathbb{R}[x]$, the set of polynomials divisible by x , denoted by (x) , is an ideal.

$$(x) = \{x \cdot f(x) \mid f(x) \in \mathbb{R}[x]\} \text{ is an ideal of } \mathbb{R}[x]$$



5. Prime Ideal

Definition:

An ideal P in a ring R is a *prime ideal* if $P \neq R$ and whenever $a \cdot b \in P$ for $a, b \in R$, then $a \in P$ or $b \in P$.

Examples:

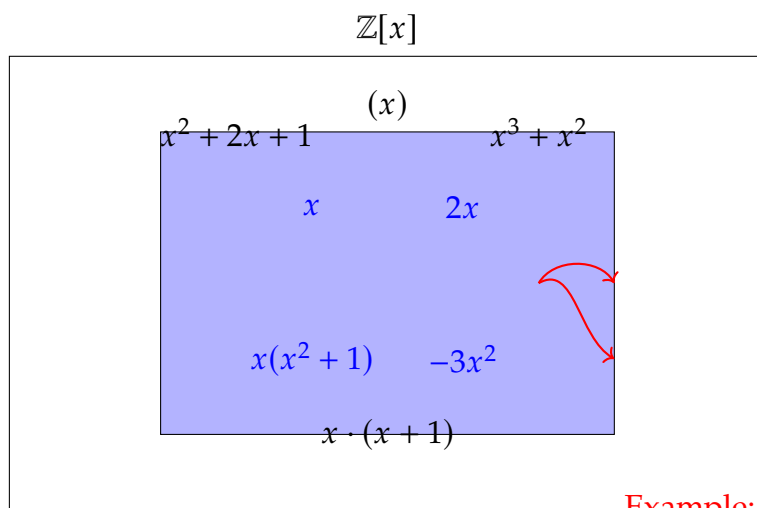
1. In $\mathbb{Z}$, the ideal (p) where p is a prime number (e.g., (5)) is a prime ideal.

$$(5) = \{5k \mid k \in \mathbb{Z}\} \text{ is a prime ideal of } \mathbb{Z}$$



2. In $\mathbb{Z}[x]$, the ideal (x) is a prime ideal.

$$(x) = \{x \cdot f(x) \mid f(x) \in \mathbb{Z}[x]\} \text{ is a prime ideal of } \mathbb{Z}[x]$$



Example:
 $(x + 1) \cdot x = x^2 + x$
 $(x^2 + x) \in (x)$ implies
 $(x + 1) \in \mathbb{Z}[x]$ or $x \in (x)$

6. Maximal Ideal

Definition:

An ideal M in a ring R is a *maximal ideal* if $M \neq R$ and there are no other ideals I such that $M \subset I \subset R$.

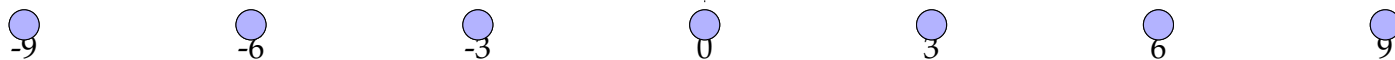
Examples:

1. In $\mathbb{Z}$, the ideal (p) where p is a prime number (e.g., (3)) is a maximal ideal.

$(3) = \{3k \mid k \in \mathbb{Z}\}$ is a maximal ideal of $\mathbb{Z}$

No larger ideal between (3) and $\mathbb{Z}$

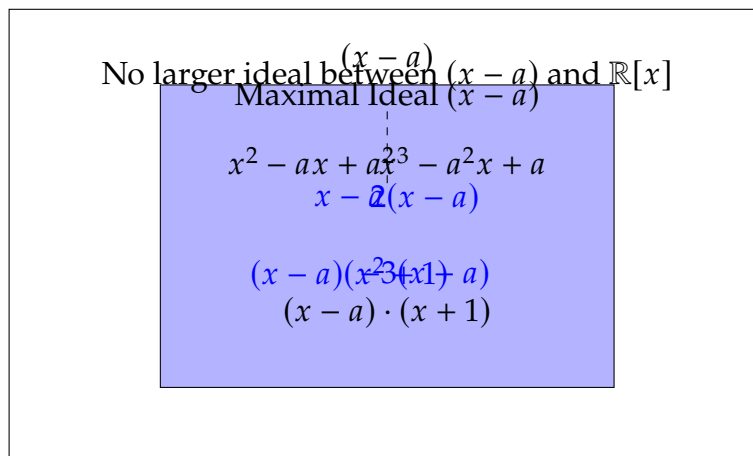
Maximal Ideal (3)



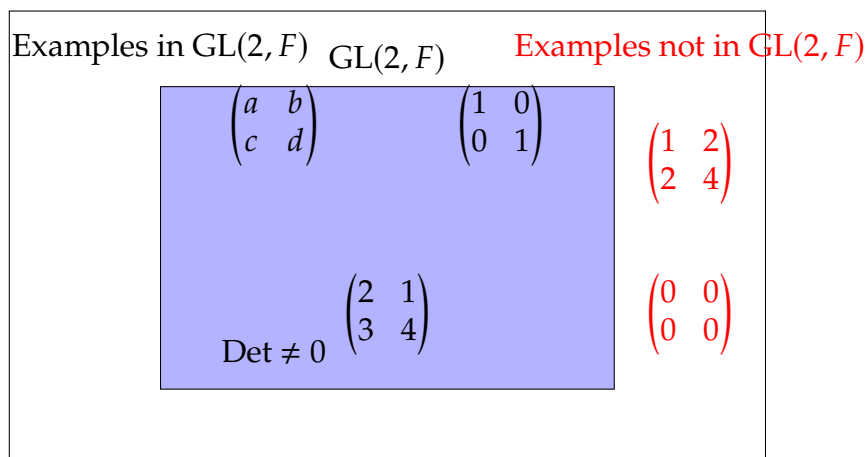
2. In $\mathbb{R}[x]$, the ideal $(x - a)$ where $a \in \mathbb{R}$ is a maximal ideal.

$(x - a) = \{(x - a) \cdot f(x) \mid f(x) \in \mathbb{R}[x]\}$ is a maximal ideal of $\mathbb{R}[x]$

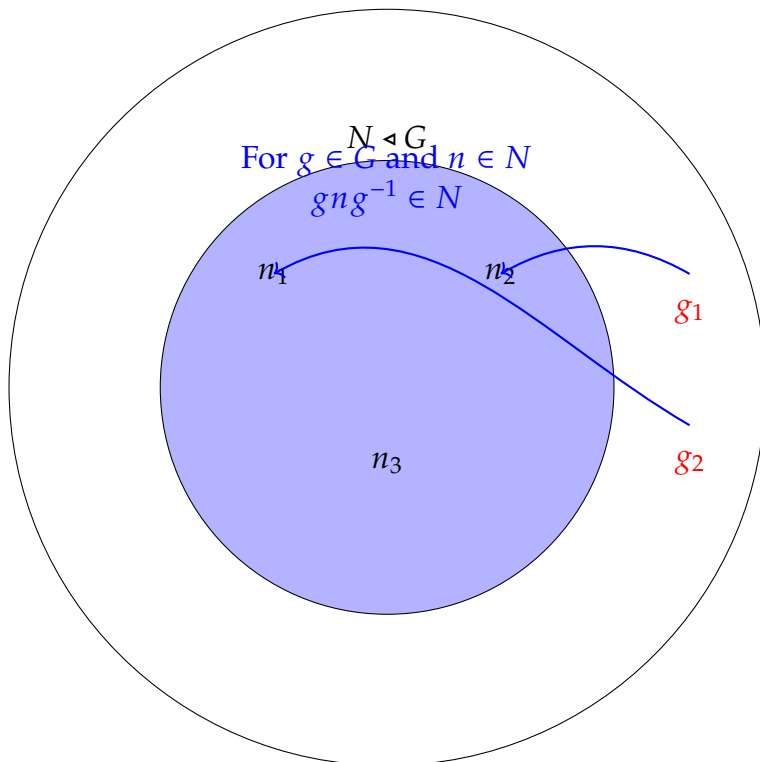
$$\mathbb{R}[x]$$



$$M_2(F)$$



$$G$$



Chapter 2

A Mapping $\mathbb{C}[x] \rightarrow \mathbb{C}$

2.1 Observation

Mapping	$\psi_p : \mathbb{Z} \longrightarrow \mathbb{Z}_p$ $n \longmapsto n \bmod p = \psi_p(n)$	$\phi_a : \mathbb{C}[x] \longrightarrow \mathbb{C}$ $f(x) \longmapsto f(a) = \phi_a(f(x))$
Additive Homo.	$\psi_p(a + b) := (a + b) \bmod p$	$\phi_a(f + g) := f(a) + g(a)$
Multiplicative Homo.	$\psi_p(ab) := (ab) \bmod p$	$\phi_a(fg) := f(a)g(a)$
Kernel	$\ker(\psi_p) = p\mathbb{Z}$	$\ker(\phi_a) = (x - a)\mathbb{C}[x]$
Image	$\mathbb{Z}_p$	$\mathbb{C}$
Ideal	$p\mathbb{Z} = \langle p \rangle$	$(x - a)\mathbb{C}[x] = \langle x - a \rangle$
Prime Ideal	$\langle p \rangle$ is prime	$\langle x - a \rangle$ is prime
Maximal Ideal	$\langle p \rangle$ is maximal	$\langle x - a \rangle$ is maximal
Isomorphism	$\mathbb{Z}_p \simeq \mathbb{Z}/p\mathbb{Z}$	$\mathbb{C} \simeq \mathbb{C}[x]/\langle x - a \rangle$

Theorem 2.1.1 Every ideal I of ring R is the kernel of some ring homomorphism.

Proof. Let I is an ideal in a ring R , that is, I is a subset of R that satisfies:

-

□

Non-Commutative Rings Without Unity and Prime Ideals

Example 1: The Ring of 2x2 Upper Triangular Matrices Over a Field $\mathbb{F}$

Consider the ring R of 2x2 upper triangular matrices over a field $\mathbb{F}$:

$$R = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in \mathbb{F} \right\}$$

This ring is non-commutative and does not have a unity element.

Prime Ideal

An ideal P in R can be the set of matrices where the (1,2)-entry is zero:

$$P = \left\{ \begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix} \mid a, c \in \mathbb{F} \right\}$$

To see why P is a prime ideal, consider matrices A and B in R . If the product $AB \in P$, then the (1,2)-entry of AB must be zero. This means either the (1,2)-entry of A is zero or the (1,2)-entry of B is zero, implying $A \in P$ or $B \in P$. Thus, P is a prime ideal.

Example 2: The Ring of Polynomials in Two Non-commuting Variables Over a Field $\mathbb{F}$

Consider the ring $R = \mathbb{F}\langle x, y \rangle$, the ring of polynomials in two non-commuting variables x and y over a field $\mathbb{F}$.

Prime Ideal

An ideal P in R can be generated by the commutator $[x, y] = xy - yx$:

$$P = (xy - yx)$$

To see why P is a prime ideal, consider two polynomials f and g in R . If $fg \in P$, then fg can be written as a multiple of $xy - yx$. If $xy - yx$ divides fg , then either $xy - yx$ divides f or $xy - yx$ divides g , meaning $f \in P$ or $g \in P$. Thus, P is a prime ideal.

Proof: Existence of a Basis for Any Vector Space using Zorn's Lemma

Theorem: Every vector space V over a field F has a basis.

Proof: To prove this, we will use Zorn's Lemma, which states:

If every chain (totally ordered subset) of a partially ordered set S has an upper bound in S , then S contains at least one maximal element.

1. **Set Construction:** Define the set C to be the collection of all linearly independent subsets of V :

$$C = \{S \subseteq V \mid S \text{ is linearly independent}\}$$

2. **Partial Order:** Partially order C by inclusion: $S \leq T$ if $S \subseteq T$.
3. **Chain:** Let $\mathcal{A}$ be a chain in C . This means that every pair of elements in $\mathcal{A}$ is comparable under $\subseteq$. For each pair $S, T \in \mathcal{A}$, either $S \subseteq T$ or $T \subseteq S$.
4. **Upper Bound for Chain:** Define $U = \bigcup_{S \in \mathcal{A}} S$. We claim that $U \in C$, i.e., U is a linearly independent subset of V .

- Suppose for contradiction that U is not linearly independent. Then there exists a finite subset $\{u_1, u_2, \dots, u_n\} \subseteq U$ and scalars $a_1, a_2, \dots, a_n \in F$, not all zero, such that:

$$a_1 u_1 + a_2 u_2 + \dots + a_n u_n = 0$$

- Each u_i belongs to some $S_i \in \mathcal{A}$, and since $\mathcal{A}$ is a chain, there exists an $S \in \mathcal{A}$ such that $u_1, u_2, \dots, u_n \in S$.
- But S is linearly independent, so the only solution to this linear combination is $a_1 = a_2 = \dots = a_n = 0$, a contradiction.

Thus, U must be linearly independent, and $U \in C$ is an upper bound for the chain $\mathcal{A}$.

5. **Application of Zorn's Lemma:** By Zorn's Lemma, C contains a maximal element B .
6. **Maximal Linearly Independent Set:** Let B be a maximal element in C . We claim that B is a basis for V .
 - B is linearly independent by construction.
 - Suppose B is not a basis for V . Then there exists some vector $v \in V \setminus \text{span}(B)$. The set $B \cup \{v\}$ would then be linearly independent (otherwise v would be in the span of B), contradicting the maximality of B .
7. **Conclusion:** Therefore, B must span V , making B a basis for V .

Proof: Co-Finite Topology is a Topology

Definition: The co-finite topology on a set X is defined as follows: a subset $U \subseteq X$ is open if and only if $U = \emptyset$ or $X \setminus U$ is finite.

Proof: We need to show that the co-finite topology on X satisfies the three properties of a topology:

1. X and $\emptyset$ are in the topology.
2. The topology is closed under arbitrary unions.
3. The topology is closed under finite intersections.

1. X and $\emptyset$ are in the topology

- X is open because $X \setminus X = \emptyset$, which is finite.
- $\emptyset$ is open by definition.

2. Closed under Arbitrary Unions

Let $\{U_i\}_{i \in I}$ be an arbitrary collection of open sets in the co-finite topology. We need to show that $U = \bigcup_{i \in I} U_i$ is open.

- If any $U_i = X$, then $U = X$, which is open.
- If all $U_i \neq X$, then $X \setminus U_i$ is finite for each i .

Consider $U = \bigcup_{i \in I} U_i$. Then:

$$X \setminus U = X \setminus \left(\bigcup_{i \in I} U_i \right) = \bigcap_{i \in I} (X \setminus U_i).$$

Since each $X \setminus U_i$ is finite, the intersection of any collection of finite sets is finite (or empty). Therefore, $X \setminus U$ is finite, which implies U is open in the co-finite topology.

3. Closed under Finite Intersections

Let $U_1, U_2, \dots, U_n$ be a finite collection of open sets in the co-finite topology. We need to show that $U = \bigcap_{i=1}^n U_i$ is open.

- If any $U_i = X$, then $U = \bigcap_{i=1}^n U_i = \bigcap_{j \neq i} U_j$, reducing the problem to fewer sets.
- If all $U_i \neq X$, then each $X \setminus U_i$ is finite.

Consider $U = \bigcap_{i=1}^n U_i$. Then:

$$X \setminus U = X \setminus \left(\bigcap_{i=1}^n U_i \right) = \bigcup_{i=1}^n (X \setminus U_i).$$

Since each $X \setminus U_i$ is finite, the union of a finite number of finite sets is finite. Therefore, $X \setminus U$ is finite, which implies U is open in the co-finite topology.

Conclusion: Since the co-finite topology on X satisfies the three properties required for a topology, it is indeed a topology.

Proof: Zero Sets of Polynomials Form a Co-Finite Topology

Definition: The zero set of a polynomial $p \in \mathbb{C}[x]$ is $Z(p) = \{z \in \mathbb{C} \mid p(z) = 0\}$.

Proof: We need to show that the collection of zero sets of polynomials in $\mathbb{C}$ forms the closed sets of the co-finite topology on $\mathbb{C}$.

1. Zero Sets are Closed in the Co-Finite Topology

Let $p(x) \in \mathbb{C}[x]$ be a non-constant polynomial of degree n . By the Fundamental Theorem of Algebra, $p(x)$ has at most n roots in $\mathbb{C}$. Therefore, $Z(p)$ is finite.

The zero set of the constant polynomial $p(x) = 0$ is $\mathbb{C}$, which is the entire set.

2. Co-Finite Topology Closed Sets

The closed sets in the co-finite topology are those whose complements are finite. That is, a set $C \subseteq \mathbb{C}$ is closed if $\mathbb{C} \setminus C$ is finite or $C = \mathbb{C}$.

3. Correspondence of Zero Sets and Closed Sets

- The zero set of a non-constant polynomial is finite, corresponding to finite closed sets in the co-finite topology.
- The zero set of the constant polynomial $p(x) = 0$ is $\mathbb{C}$, corresponding to $\mathbb{C}$ itself being a closed set in the co-finite topology.

4. Characterization of Closed Sets

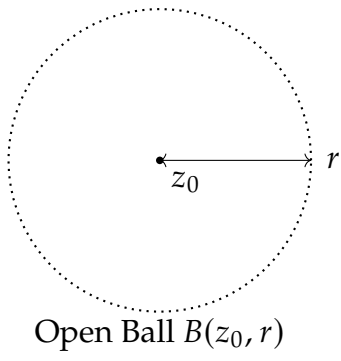
To show that the closed sets in the co-finite topology are precisely the zero sets of polynomials, consider:

- A finite set $\{z_1, z_2, \dots, z_n\}$. This can be written as the zero set of the polynomial

$$p(x) = (x - z_1)(x - z_2) \cdots (x - z_n).$$

- The whole set $\mathbb{C}$ is the zero set of the polynomial $p(x) = 0$.

Conclusion: The closed sets in the co-finite topology on $\mathbb{C}$ are exactly the zero sets of polynomials in $\mathbb{C}[x]$. Therefore, the collection of zero sets of polynomials forms the closed sets of the co-finite topology, showing that the zero sets of polynomials generate the co-finite topology on $\mathbb{C}$.



Hence, every vector space V over a field F has a basis, as guaranteed by Zorn’s Lemma.

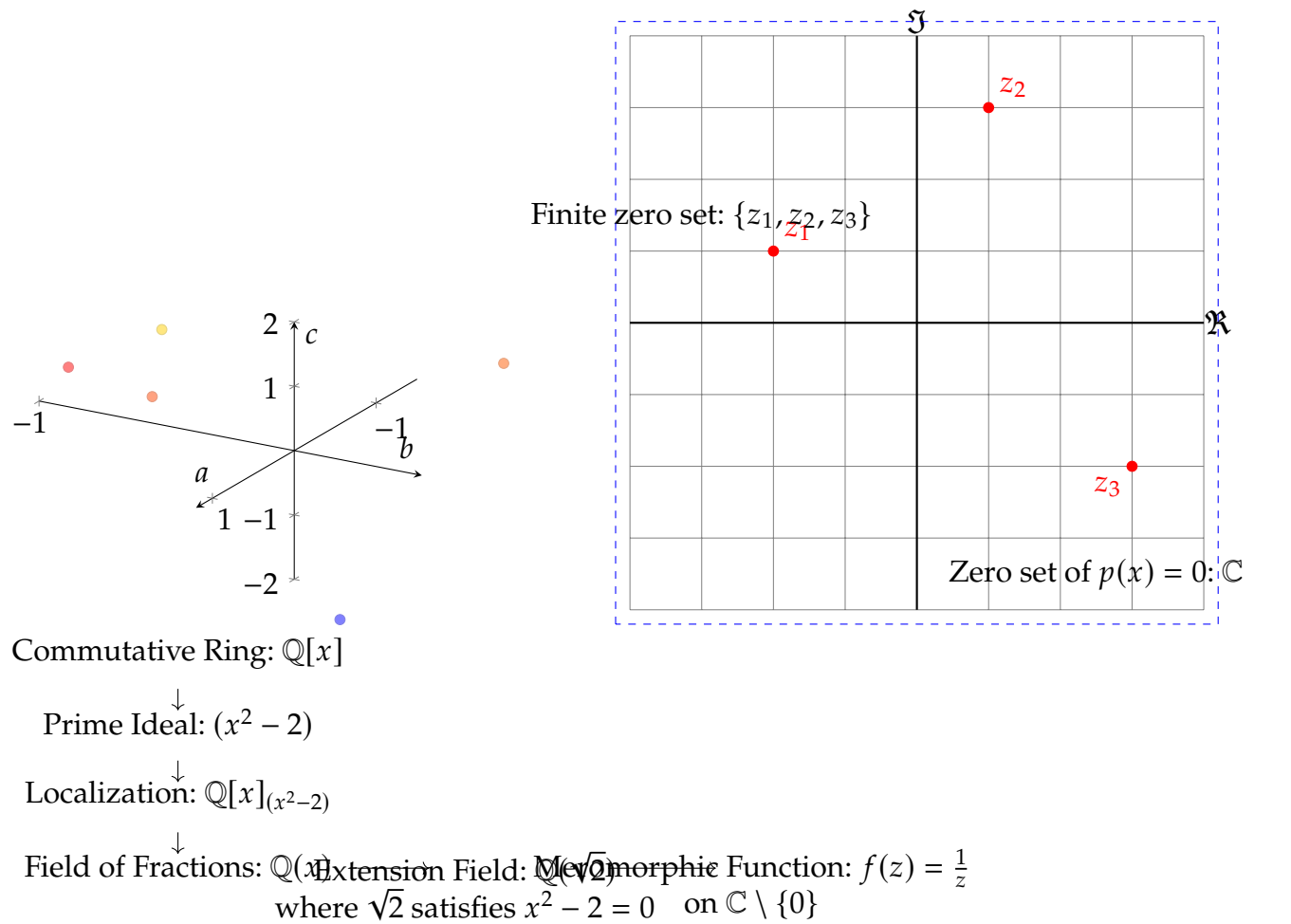
$$\mathbb{C}[x]/\langle x-\alpha \rangle = \mathbb{C}[x]/(x-\alpha)\mathbb{C}[x] \simeq \mathbb{C}.$$

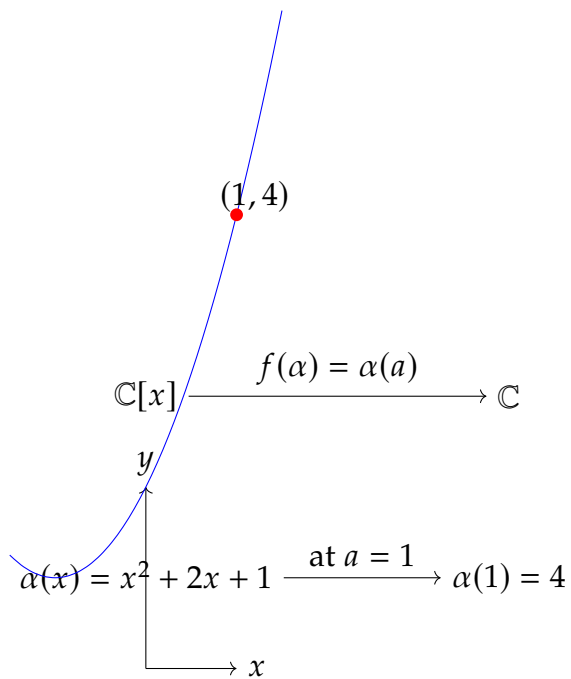
Let $f, g \in I = \mathbb{C}[x]/(x-\alpha)\mathbb{C}[x]$. Then

$$f(x) = (x-\alpha)q_1(x) + r(x), \quad g(x) = (x-\alpha)q_2(x) + r(x)$$

Then

$$f(x) \equiv g(x) \pmod{(x-\alpha)\mathbb{C}[x]} \iff f(x) - g(x) \in (x-\alpha)\mathbb{C}[x]$$





$$\mathbb{C}[x] \xrightarrow{f(\alpha) = \text{leading coefficient}} \mathbb{C}$$

$$\alpha(x) = 3x^2 + 2x + 1 \xrightarrow{\text{leading coefficient}} 3$$

$3x^2$

$$\mathbb{C}[x] \xrightarrow{f(\alpha) = \text{constant term}} \mathbb{C}$$

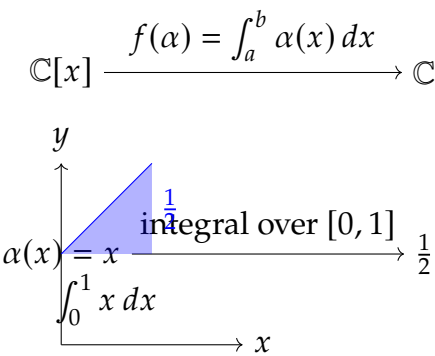
$$\alpha(x) = x^2 + 2x + 1 \xrightarrow{\text{constant term}} 1$$

constant term

$$\mathbb{C}[x] \xrightarrow{f(\alpha) = \sum a_i} \mathbb{C}$$

$$\alpha(x) = x^2 + 2x + 1 \xrightarrow{\text{sum of coefficients}} 4$$

$$1 + 2 + 1 = 4$$

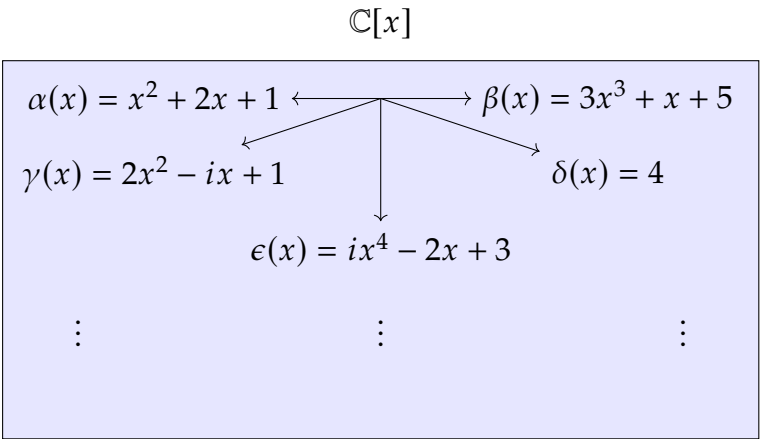
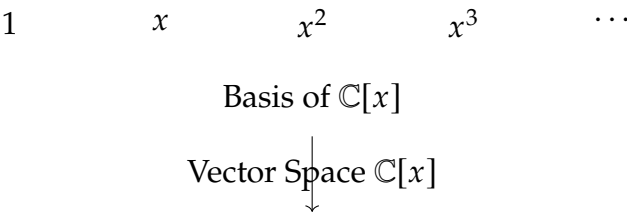


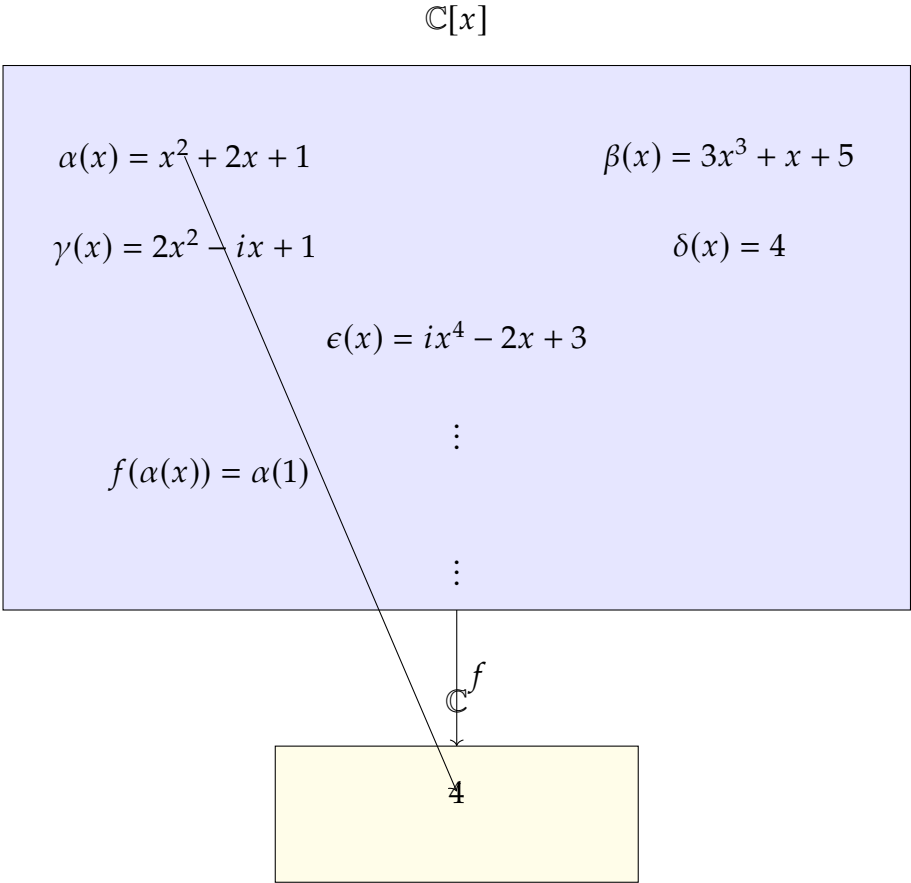
$$\mathbb{C}[x]$$

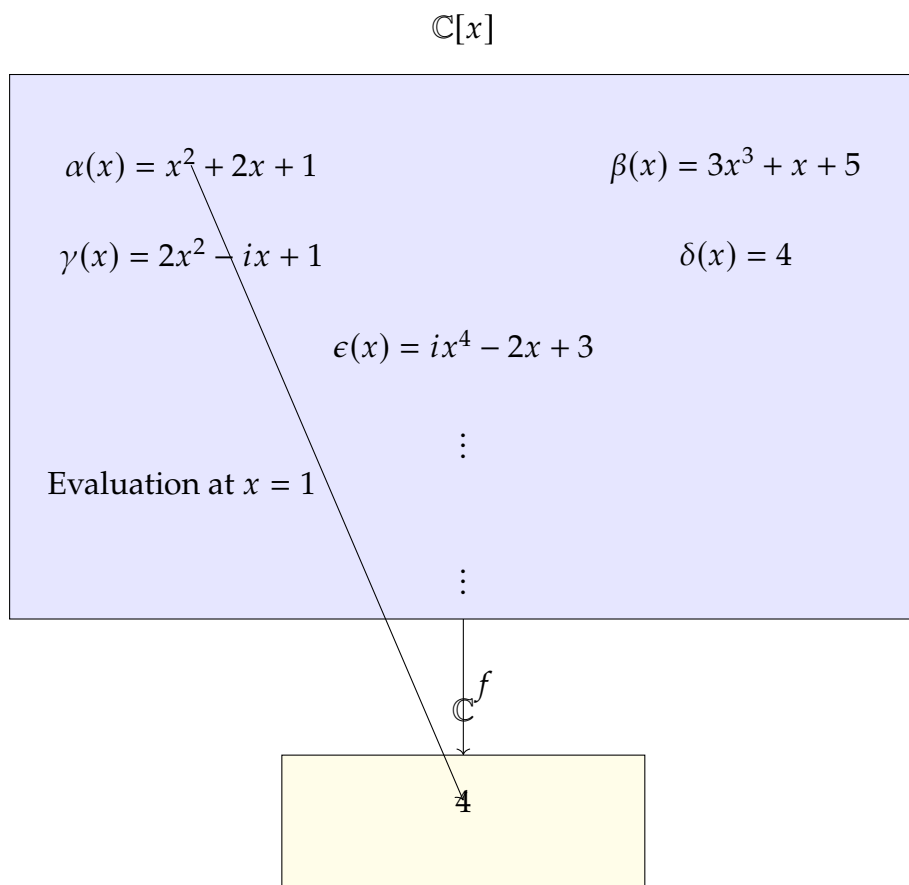
$$\begin{aligned} \alpha(x) &= x^2 + 2x + 1 & \beta(x) &= 3x^3 + x + 5 \\ \gamma(x) &= 2x^2 - ix + 1 & \delta(x) &= 4 \\ \epsilon(x) &= ix^4 - 2x + 3 \end{aligned}$$

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$$

General Form



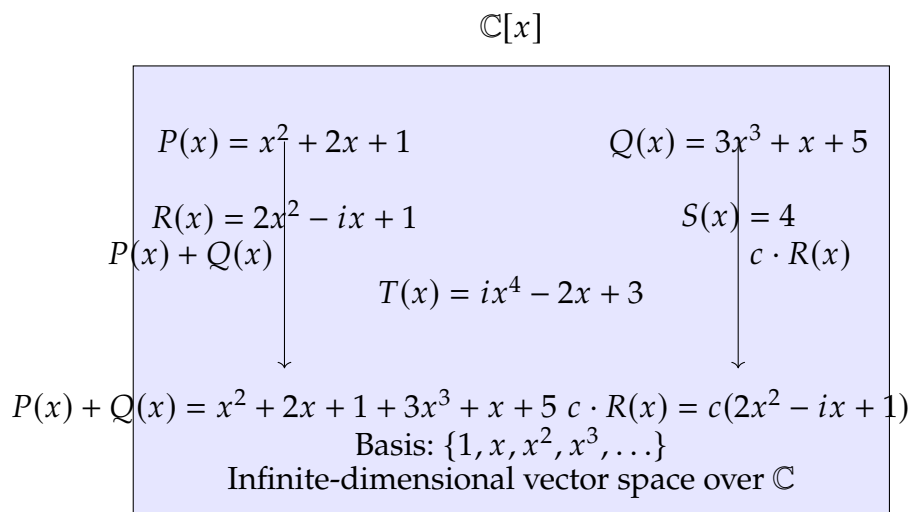




Leading Coefficient: 3

Constant Term: 1

Sum of Coefficients: 4



2.2 Zorn's Lemma and Basis

2.2.1 Relations

Let $\mathcal{R} \subseteq A \times A$ be a relation on a set A .

Property Type	Condition	Definition	Example
Reflexive Properties	Reflexive	$\forall a \in A, (a, a) \in R$	Equality relation on $\mathbb{R}$
	Coreflexive	$\forall a, b \in A, (a, b) \in R \Rightarrow a = b$	Identity relation $\{(a, a) : a \in A\}$
	Irreflexive (Anti-reflexive)	$\forall a \in A, (a, a) \notin R$	"Is less than" relation on $\mathbb{R}$
	Non-Reflexive	$\exists a \in A, (a, a) \notin R$	"Is a friend of" relation
Symmetric Properties	Symmetric	$\forall a, b \in A, (a, b) \in R \Rightarrow (b, a) \in R$	Equality relation on $\mathbb{R}$
	Asymmetric	$\forall a, b \in A, (a, b) \in R \Rightarrow (b, a) \notin R$	"Is parent of" relation
	Anti-Symmetric	$\forall a, b \in A, (a, b) \in R \text{ and } (b, a) \in R \Rightarrow a = b$	"Is less than or equal to" relation on $\mathbb{R}$
	Non-Symmetric	$\exists a, b \in A, (a, b) \in R \text{ and } (b, a) \notin R \text{ or } (a, b) \notin R \text{ and } (b, a) \in R$	"Is sibling of" relation
Transitive Properties	Transitive	$\forall a, b, c \in A, (a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R$	Divisibility relation on $\mathbb{N}$
	Anti-Transitive	$\forall a, b, c \in A, (a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \notin R$	Hard to find simple example
	Non-Transitive	$\exists a, b, c \in A, (a, b) \in R \text{ and } (b, c) \in R \text{ and } (a, c) \notin R$	"Is a friend of"

Table 2.1: Conditions and Examples for Relation Properties

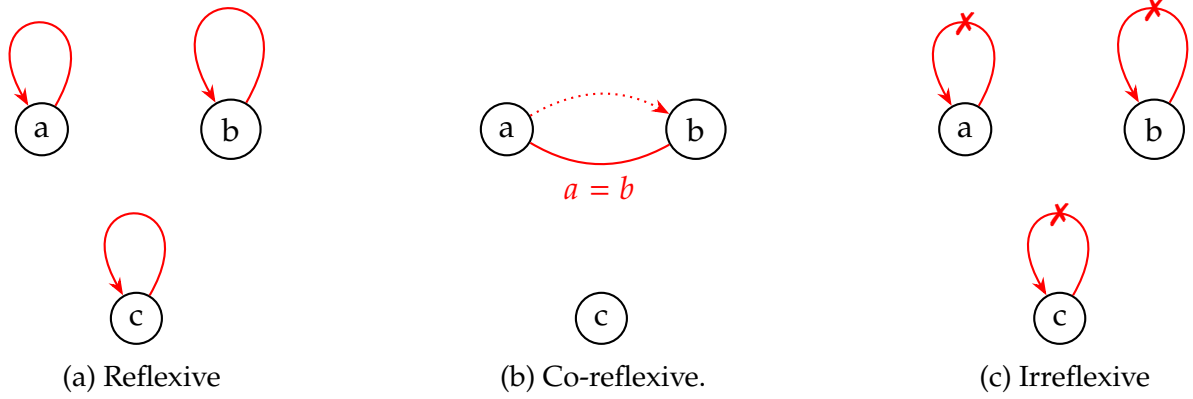


Figure 2.1: Reflexivity

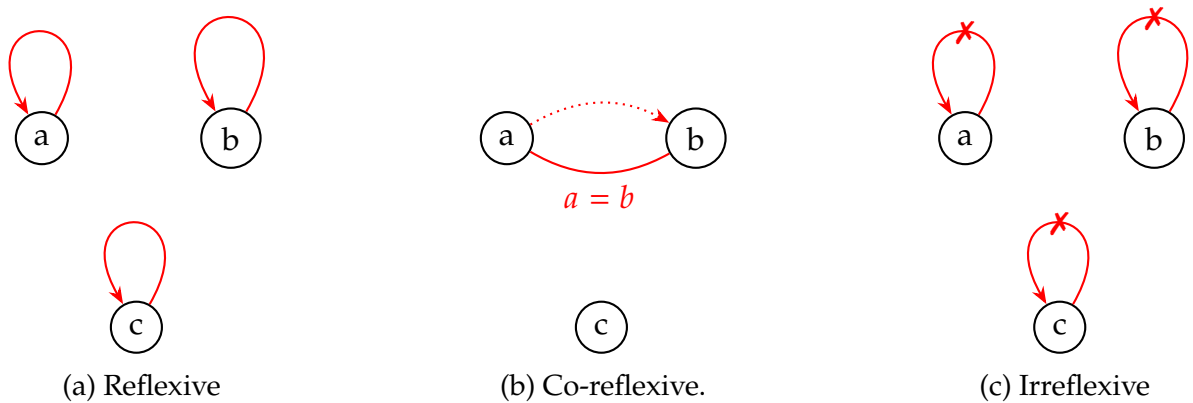
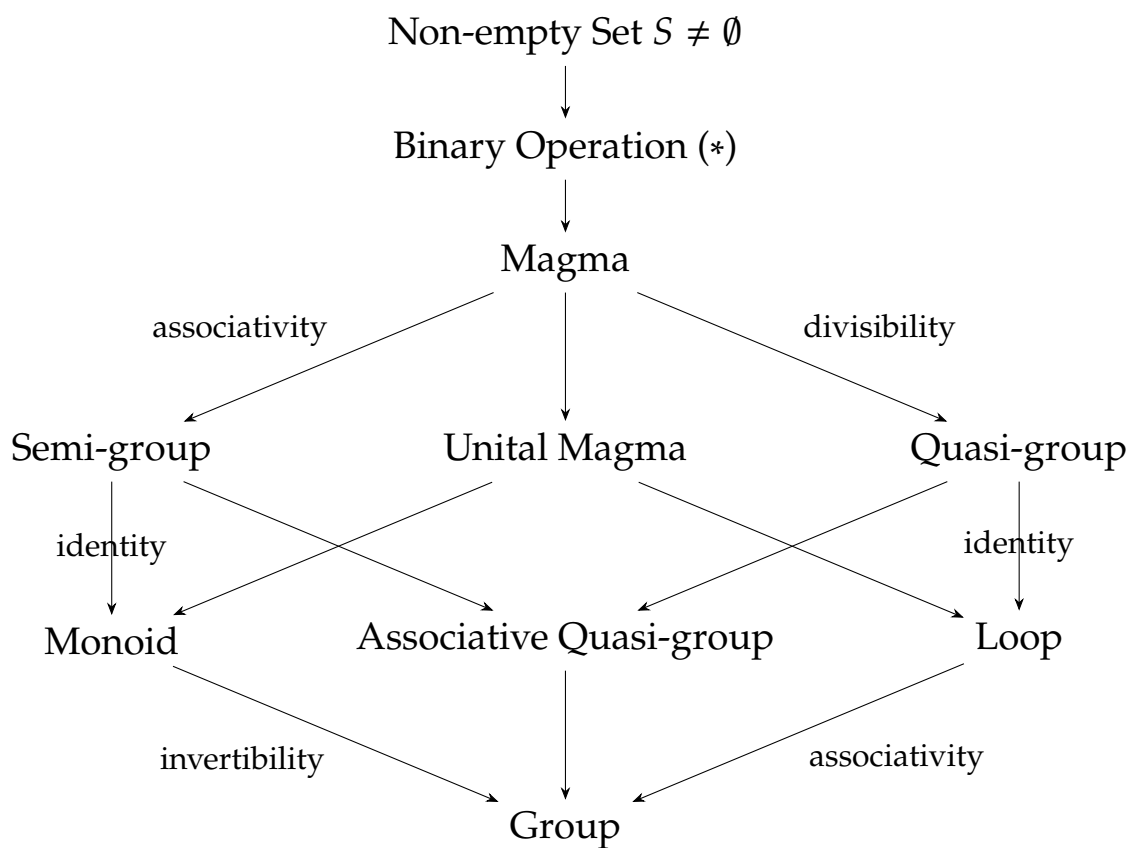


Figure 2.2: Reflexivity



Chapter 3

Localization

Observation. Consider the ring of integers $(\mathbb{Z}, +, \times)$. For $n \neq 1$, consider the set of integer multiples $n\mathbb{Z}$. Then $1 \notin n\mathbb{Z}$, but $(n\mathbb{Z}, \times)$ is closed under $\times$. That is, $n\mathbb{Z}$ is closed under $\times$ but not actually multiplicatively closed.

Multiplicatively Closed Subset of Ring

Definition 3.1. Let $(R, +, \times)$ be a ring with unity 1_S and zero 0_S . Then $S \subseteq R$ is **multiplicatively closed** iff

- $1_R \in S$.
- $x, y \in S \implies x * y \in S$.

Note (From $\mathbb{Z}$ to $\mathbb{Q}$).

- (Numerator) $R = \mathbb{Z}$ is a ring
- (Denominators) $S = \mathbb{Z} \setminus \{0\}$ is multiplicatively closed and $1 \in S$
- ($\mathbb{Q} \simeq S^{-1}R$) Every $q \in \mathbb{Q}$ is of the form $s^{-1}r$ for $r \in R$ and $s \in S$.
- (Equivalence Relation)

$$\frac{a}{b} = \frac{c}{d} \iff ad = bc \iff t(ab - bc) = 0 \text{ for } t \in S.$$

- $\mathbb{Q}$ is ring:
 - $\mathbb{Q}$ has an addition $a/b + c/d = (da + bc)/bd$
 - $\mathbb{Q}$ has a multiplication $a/b \cdot c/d = ac/bd$
 - $\mathbb{Q}$ has a zero $0/1$ and a one $1/1$.
- $\mathbb{Z}$ is a subring of $\mathbb{Q}$ by $r \mapsto r/1$.

Note (Functions close to 0 - "local functions").

- (Numerator) $R = \mathbb{R}[x] \in \mathbb{R}^{\mathbb{R}}$ is a ring
- (Denominators) $S = \{s \in R : s(0) \neq 0\}$ is multiplicatively closed and $1 \in S$
- ($L \simeq S^{-1}R$) Every local function L is of the form $s^{-1}r$ for $r \in R$ and $s \in S$.

- (Equivalence Relation)

$$\frac{a}{b} = \frac{c}{d} \iff ad = bc \iff t(ab - bc) = 0 \text{ for } t \in S.$$

- Local function is ring:
 - L has an addition $a/b + c/d = (da + bc)/bd$
 - L has a multiplication $a/b \cdot c/d = ac/bd$
 - L has a zero $0/1$ and a one $1/1$.
- R has a map to L by $r \mapsto r/1$.

3.1 For completeness: The formal definition/statement

Let R be a commutative ring and S be a multiplicatively closed set with $1 \in S$.

- (a) Equivalence Relation on $R \times S$

$$(a, b) \sim (c, d) \iff \exists t \in S : t(ad - bc) = 0.$$

- (b) [Localization (localize at S)] The set of equivalence classes $S^{-1}R$.

- (c) Addition on $S^{-1}R$

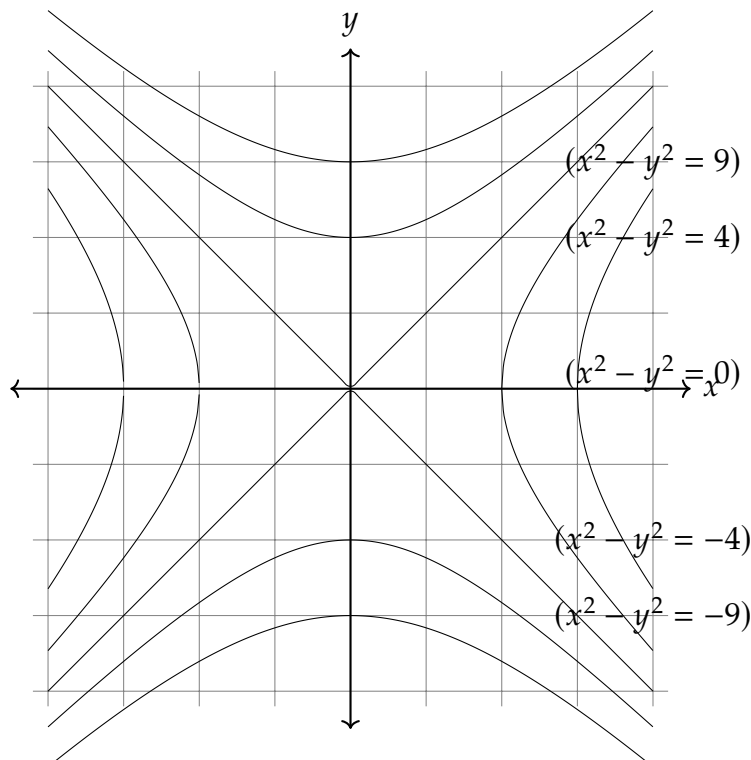
$$(a, b) + (c, d) = (ad + bc, bd)$$

- (d) Multiplication on $S^{-1}R$

$$(a, b) \cdot (c, d) = (ac, bd)$$

- (e) (Slogan. Invert elements of S)

- $S^{-1}R$ is a ring with zero $(0, 1)$ and one $(1, 1)$
- There is a ring homomorphism $\phi : R \rightarrow S^{-1}R$ given by $r \mapsto (r, 1)$, i.e., $r \mapsto r/1$
- ϕ is injective (R is a subring of $S^{-1}R$) iff S contains no zero divisor



- $X \subsetneq \mathbb{R}^2$ is the vanishing set of $(x - y)(x + y)$
- Model: the ring $R = \mathbb{R}[x, y]/\langle (x - y)(x + y) \rangle$
- $\langle x - y \rangle : X \rightarrow \mathbb{R}$ is locally at a indistinguishable from 0
- $\langle x - y \rangle - 0 \neq 0$ but $(x + y)\langle x - y - 0 \rangle = 0$ in R
- So $\langle x - y \rangle = 0$ in R localized at $S = \{s \in R : s(a) = 0\}$.